

Complete Genome Sequence of Cluster A1 *Mycobacterium smegmatis* Bacteriophage Arlo

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ABSTRACT Mycobacteriophage Arlo is a newly isolated *Siphoviridae* bacteriophage isolated from soil samples collected in Bluff Dale, Texas. Mycobacteriophage Arlo has a 52,960 base-pair double-stranded DNA genome that is predicted to contain 96 protein-coding genes. Mycobacteriophage Arlo is related to mycobacteriophage DD5 and other cluster A1 bacteriophages.

We report the whole-genome sequence of mycobacteriophage Arlo (1, 2), which was directly isolated from a strain of *Mycobacterium smegmatis*, a rapidly growing environmental species that is generally nonpathogenic but can act as an opportunistic pathogen in immunosuppressed individuals (3). Mycobacteriophage Arlo was isolated from compost-containing community vegetable garden soil samples collected in Bluff Dale, Texas (32°19'08.0004", -098°01'14.9016"). Soil samples were washed with 7H9 liquid medium, and bacteriophages were extracted from the mixture through a 0.22-μm filter. For virus replication, filtered medium was incubated with *Mycobacterium smegmatis* mc²155 at 37°C for 48 h. Plaque assays of isolated mycobacteriophage Arlo resulted in medium-sized turbid plaques. Negative-staining transmission electron microscopy showed that mycobacteriophage Arlo has a siphoviral morphology with a 60-nm-diameter nonenveloped icosahedral capsid and a 125 nm flexible noncontractile tail, which is typical of viruses in the *Caudovirales* order (Fig. 1).

DNA was isolated from purified bacteriophage with the Promega Wizard DNA clean-up kit, and sequencing libraries were prepared from genomic DNA with the NEBNext Ultra II kit. Libraries were sequenced with Illumina MiSeq at the Pittsburgh Bacteriophage Institute to approximately 1,987-fold coverage from 742,500 total reads of 150-base read length (4). Sequence reads were assembled with Newbler 2.9 with default settings to produce a single-bacteriophage contig, which was checked for completeness, accuracy, and genome termini using *conseq* v29.0 (5). The virus was determined to contain a linear double-stranded DNA genome that is 52,960 base pairs in length, with 63.8% G+C content, and a 3' single-stranded terminal overhang of 5'-CGGATGGTAA-3'. Whole-genome nucleotide alignment with NCBI BLASTn (<https://blast.ncbi.nlm.nih.gov/>) (6) showed 96–97% nucleotide identity to cluster A1 mycobacteriophages Oogway (GenBank accession number [MH230878](#)) and DD5 (GenBank accession number [NC_011022](#)) (2).

Autoannotation of the genome was performed using GLIMMER v3.02 (7, 8) and GeneMark v2.5p (9, 10), followed by manual inspection, refinement of start sites, and annotation revision using Phamerator (<https://phamerator.org/>) (11), DNA Master v5.23.2 (<http://phagesdb.org/DNAMaster/>), and PECAAN (<https://pecaan.kbrinsgd.org/>). Mycobacteriophage Arlo is predicted to contain 96 protein-coding genes. No tRNAs genes were identified by ARAGORN v1.2.38 (12) or tRNAscan-SE v2.0 (13). Start codon usage was determined to be 90.12% AUG, 8.72% GUG, and 1.16% UUG.

HHpred v3.0beta (14, 15) and NCBI BLASTp (6) software were used to assign putative

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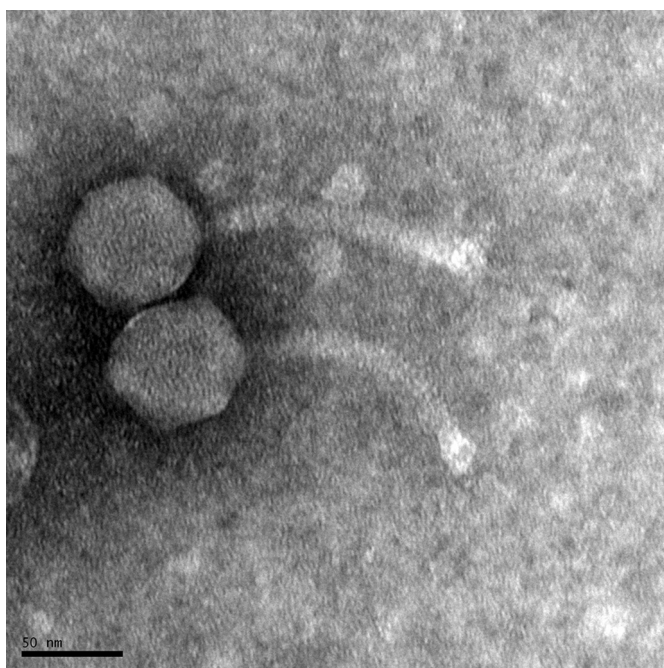


FIG 1 Transmission electron microscopy (TEM) of mycobacteriophage Arlo. Purified high-titer lysate was placed on a carbon type-B 300 mesh grid, stained with uranyl acetate, and imaged with an FEI Tecnai G² Spirit BioTWIN transmission electron microscope (NL1.160G). TEM micrographs of negatively stained mycobacteriophage Arlo show an approximately 60-nm-diameter nonenveloped icosahedral capsid and 125-nm flexible noncontractile tail. The morphology of mycobacteriophage Arlo corresponds to that of members of the *Siphoviridae* family.

functions to 34 (35.4%) of 96 predicted protein-coding genes. The mycobacteriophage Arlo genome is arranged with rightwards-transcribed genes (genes 1 to 36, 58.4% of genome) encoding virion structural and assembly proteins and a lysis cassette consisting of lysin A and lysin B genes. Leftwards-transcribed genes encode DNA polymerase I, metallophosphoesterase, DNA primase, DNA methylase, endonuclease VII, NrdH-like glutaredoxin, DnaB-like dsDNA helicase, RecB-like exonuclease/helicase, and immunity repressor proteins.

Data availability. The mycobacteriophage Arlo genome is available at GenBank as accession number [MH576971](https://www.ncbi.nlm.nih.gov/nuclseq/MH576971). Raw reads are available in the SRA under accession number [SRX4721440](https://www.ncbi.nlm.nih.gov/sra/SRX4721440).

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